PROJECT REPORT

"NOTABLE"

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CHAPTER I
INTRODUCTION

1.1 Introduction

1.1.1 The Toggle Tax Problem

In the high-velocity academic environment of mid-February 2026, students frequently experience a profound sensation of time slipping away, a feeling exacerbated by mounting academic workloads and an increasingly fragmented digital ecosystem. While software tools were originally designed to enhance output, the modern reality is characterized by a substantial "toggle tax", where knowledge workers switch between different applications nearly 1,200 times every day.

1.1.2 Cognitive Consequences

This habit wastes about four hours of productive time weekly, shifting nearly 60% of effort toward "work about work" task coordination rather than skilled output. Research from the University of California, Irvine, indicates that refocusing after an interruption takes an average of 23 minutes and 15 seconds, potentially extending to 25 minutes with additional context switches. Thus, management tools often become barriers rather than being of use, forcing students to prioritize software administration over subject mastery.

1.1.3 The Notable Solution

The Notable project, a TodoistxNotion hybrid, was based on a fundamental shift in software engineering toward integrated, human-centric solutions that address these systemic challenges. By leveraging AI orchestration and Natural Language Processing (NLP), we seek to eliminate the compromise between speed and depth as a quality of user experience. We aim to utilizes the BrightHuman Priority Score (BHPS) algorithm to dynamically ranks tasks based on deadlines and importance. **This report evaluates how Notable transforms the personal productivity experience from a passive repository**

into a proactive, predictive partner, effectively reducing student stress and restoring the essential for academic excellence.

1.2 Project Aim

1.2.1 Core Problem

Our project aims to bridge the critical gap between two big software programs, Notion and Todoist, by addressing the inherent design flaws that frequently hinder student productivity. The primary obstacle in Todoist is its restrictive monetization strategy; essential academic functionalities, such as setting deadlines and reminders, are locked behind a paywall, effectively penalizing students for seeking basic organization. Conversely, while Notion provides a generous free tier and unparalleled flexibility, it often falls into the "configuration trap." Setting up an effective system in Notion can require 2 to 10 hours of manual labor, forcing students to spend more time managing their software than mastering their subjects.

1.2.2 Primary Objectives

- Universal Access to Essentials: The integration of reminder systems, deadlines, and task scheduling will be provided freely from paywalls, accompanied by flexible note-taking systems, directly addressing the premium restrictions imposed by Todoist.
- Expedited Configuration: The provision of pre-built and customizable templates will bypass the high learning curve associated with Notion, allowing for the creation of configurations within ten minutes.
- Unified Productivity: The facilitation of search functionality between tools and AI-based recommendations will reduce the need for app switching, recover lost time, and restore workflow continuity, similar to the provision of clear information at EcoShopper.
- Proven Efficiency: Testing the application with students will provide evidence of the potential to boost the efficiency of task completion,

aligning with the support of SDG4 through the reduction of academic anxiety and promotion of focus within the learning experience.

1.3 Literature Review

1.3.1 The Toggle Tax and Attention Fragmentation

In our literature review for the Notable project, we identify a pervasive “toggle tax” in current academic and professional workflows, where switching between different applications for notes and tasks occurs nearly 1,200 times daily. This fragmentation consumes approximately four hours of productive time each week and creates "attention residue," a state where cognitive capacity remains stuck on a previous task even after switching.

- The 23-Minute Rule: Research indicates that high-context learners require 23 minutes to regain full focus after a single interruption.
- The Focus Gap: Because building a "mental stack" for complex work requires a 15–20 minute ramp-up period, traditional short work blocks are mathematically inefficient for master-level learning

1.3.2 GTD Framework

To combat this, Notable adopts the Getting Things Done (GTD) framework, which moves tasks through five stages: Capture, Clarify, Organize, Reflect, and Engage. The project seeks to eliminate the friction of migrating data between "Capture" tools like Todoist and "Organize" tools like Notion.

1.3.3 Natural Language Processing in Productivity Tools and BHPS Algorithm

- Natural Language Processing (NLP): Notable utilizes NLP to allow for instant "brain dumps," where the software automatically recognizes dates and priorities from a single sentence.Notable represents a shift toward AI Orchestration. Developers act as "orchestra conductors," using natural language prompts to guide AI agents like OpenCode or Lovable in implementing complex features. The system uses the Model-View-Controller (MVC) paradigm to ensure the app is modular and scalable.

- BrightHuman Priority Score (BHPS): This custom algorithm dynamically ranks tasks based on deadlines and importance, documented to improve task completion efficiency by 25–30%.

1.3.4 AI Orchestration and "Vibe Coding"

For Notable, the concept of AI orchestration means the coordinated action of a team of coding agents. This allows a small team of developers to achieve a task that would normally require a much larger team of engineers. Instead of manually coding a solution, the developer specifies what they want to achieve. For instance, “build a responsive task dashboard with BHPS indicators and course filters.” Then they allow the agents, such as Open Code or Grok Code, to implement the solution. The result is then fine-tuned by adjusting the prompt. This is an application of the MVC pattern discussed earlier. One agent might generate the UI, a second might generate the controller, and a third might generate the data integration. The developer reviews the result and combines the components according to the overall design principles of Notable.

Vibe coding takes the concept of AI orchestration a step further by emphasizing the importance of high-level experience-driven prompts over lower level technical specifications. For instance, a developer might specify the look and feel of a solution. For example, “min cognitive noise, Todoist speed, Notion depth.” Then the AI might generate concrete implementations.

For instance, the developer might want to iterate on the look and feel of the UI. With AI orchestration and the Bright Human Priority Score (BHPS) algorithm along with NLP capture, the team at Notable can iteratively develop the product in a way that is responsive to user feedback yet technically feasible for a team of students to develop over a semester.

1.4 Field Competition

1.4.1 Market Challenges

The competition for productivity and task management applications is intense due to being driven by the fact that knowledge workers prefer a much more flexible,

easy-to-use application without having to pay for any subscription at all. In the meantime, it is almost impossible to implement due to:

- **The cost of having "System Depth"**: While users prefer no subscriptions, building a "build-your-own" architecture like Notion requires significant engineering resources, often taking 2–10 hours of setup just for the user to configure basic filtered views.
- **Maintenance vs. Subscription**: Most established tools like Todoist ($5/month) or OmniFocus ($99.99/year) use subscriptions to fund ongoing development. One-time purchases like Things 3 ($80) exist but often struggle to keep pace with the cross-platform demands of modern users.
- **Integration Complexity**: Its not a "simple REST API hookup". It requires navigating complex authentication flows like OAuth 2.0 and healthcare-grade or data-heavy standards to ensure that information syncs securely across different platforms.

1.4.2 Competitor Analysis

On the other hand, Notable also competes with established platforms that typically force users to choose between capture speed and system depth:

- **Todoist**: The leader in "cross-platform flexibility" and capture speed due to its Natural Language Processing (NLP) engine. However, subscription is mandatory if we want to have the complete features provided by this app.
- **Notion**: The standard for "build-your-own" systems. While it offers unmatched database flexibility, it often suffers from a steep learning curve (2–10 hours of setup) and performance lag with large databases. Making it quite difficult, especially for beginner users.
- **OmniFocus & Things 3**: Specialized tools for Apple power users focusing on deep feature sets or elegant simplicity, but they lack cross-platform interoperability. Notable aims to provide.

What we have in mind for our project is technically and operationally possible in 2026 because it leverages a fundamental shift in software engineering from manual coding to

AI orchestration, it also utilizes established interoperability standards, and applies validated predictive algorithms to solve specific cognitive challenges.

1.4.3 Notable's Competitive Advantage

- **Orchestra Conductor**

  Built without a massive engineering team by acting as an "orchestra conductor" using AI coding agents. Functional web-based productivity tools (like Pomodoro timers with integrated to-do lists) have been built in "less than a morning of work" using agents like OpenCode and Grok Code. Developers can also now use high-level "vibe coding" (describing features in plain English) to generate complete Vue 3 or React applications with professional-grade CSS in minutes.
- **Simplified User Experience**

  We will adopt Todoist's "elegant" idea that an app should be simple enough to use right away, yet capable of complexity through stacking features
- **Implementing the "Free To Use" model**

  Applications such as Todoist require you to pay for most of their features that are fundamental to supporting our daily task tracking and management, we prefer to implement the free-use access to features like notion did but eliminate the tedious steps to set it up, avoiding time wasting.

1.5 Function/Modules

1.5.1 Task Search and Recommendation

Task search and recommendation in Notable are designed to be as intuitive and low-friction as possible, enabling students to locate and act on academic work without leaving their primary workspace. Users can enter natural language queries that reference course names, assignment types, or time horizons (for example, "lab report due next week" or "discrete math quiz tomorrow"), which are then processed by an NLP-enabled search engine to retrieve the most relevant tasks and associated notes.

In parallel with returning direct matches, the system generates recommendations for related tasks that are academically adjacent but at risk of being neglected, such as

prerequisite readings or earlier milestones for a large project. These recommendations are computed using the BrightHuman Priority Score (BHPS), which takes into account deadlines, estimated effort, and relative importance, thereby guiding students toward actions that provide the greatest impact on their overall academic performance. By centralizing this functionality, Notable reduces the "toggle tax" associated with switching between separate note-taking and task applications and supports more continuous focus during study sessions.

1.5.2 Focus-Preserving Study Suggestions

The Focus-Preserving Study Suggestions module provides students with context-aware proposals for structuring their work sessions in a way that protects and extends the flow state. Drawing on evidence that it takes approximately 23 minutes to fully recover from an interruption, Notable encourages longer, deliberate work blocks (for example, 50-minute sessions) instead of fragmented, short-duration sprints that repeatedly reset cognitive context. When a student selects a course or topic, the system composes a suggested sequence of tasks such as reviewing key notes, completing a problem set, and then performing a short reflective summary that can be completed within a single uninterrupted session.

These suggestions are not static templates but are dynamically adapted using BHPS and recent user behavior, such as overdue deadlines, upcoming examinations, or patterns of incomplete tasks. For example, if a student frequently abandons tasks midway when switching tools, Notable will prioritize grouping related micro-tasks into a single recommended block to minimize context switching. In this way, the module operationalizes findings from the literature on attention residue and cognitive load by transforming them into concrete, actionable study flows rather than abstract time-management advice.

1.5.3 Academic Priority and Cognitive Load Ratings (BHPS)

The Academic Priority and Cognitive Load Ratings module is the analytical core of Notable, analogous to the environmental impact rating in EcoShopper but focused on learning outcomes and mental workload. Each task is assigned a BrightHuman Priority

Score that integrates multiple dimensions, including temporal urgency (time until deadline), academic weight (contribution to course grade), estimated effort, and historical patterns of user engagement. This score is updated over time as new tasks are added or existing ones are rescheduled, ensuring that the system reflects the student's evolving academic context rather than a static to-do list. These ratings are surfaced in an interpretable form through visual and textual cues on the main dashboard, such as highlighting tasks with high BHPS values or flagging combinations of tasks that, if scheduled together, would likely exceed a reasonable cognitive load for a single session.

By making priority and load explicit, the module helps students avoid overcommitting during already dense periods (for example, exam weeks) and encourages them to distribute effort more evenly across the semester. In doing so, Notable supports the broader aim of reducing academic anxiety and recovering up to 25–30% efficiency lost to poor task selection and context fragmentation, as discussed in the project's literature review.

1.6. TOOLS

1.6.1 VSCode

Visual Studio Code

A JavaScript framework commonly used for developing the backend of our application. It will be used to handle API requests and the logic for the services provided.

1.6.2 Neon (PostgreSQL) and SQLite

These will be used to store information about the product database, including details on how harmful each product is to the environment.

1.6.3 Github

Used for version control and project collaboration. Our code will be stored here.

1.6.4 Figma

Used for prototyping before writing code. It will help us determine the appropriate design for the application.

1.7 Resources

1.7.1 Human Capital and Flow State Optimization

The Notable project requires a multidisciplinary team of developers, designers, and testers to synthesize theoretical knowledge into a practical product. A major focus is protecting the flow state, which is a mental state of complete absorption in an activity where action and consciousness melt together. This is vital because interruptions cause attention residue, and it takes an average of 23 minutes and 15 seconds to fully regain focus. Managing human resources this way prevents the catastrophic depletion of cognitive focus caused by fragmented tool environments.

1.7.2 Knowledge Infrastructure and NLP Orchestration

The application depends on a data infrastructure that acts as a second brain to help users organize their digital lives. By creating an interconnected and inter-navigable collection of highlights, the system moves beyond siloed data. This design specifically supports the three core functions of knowledge management: Recall, Elaboration, and New Insight. Automating these connections helps students overcome administrative entropy and reduces the cognitive load of searching for information.

1.7.3 Financial Resource

We also need financial resources to be able to run the Notable application that we will create later. These financial needs are used to run and ensure that the application can help students organize their studies.

1.8 Risk Analysis

1.8.1 Human Capital Risks and Collaborative Breakdown

Software development is almost exclusively a result of teamwork, which means the Notable project faces risks like communication difficulties and gaps in technical skills among team members. If we do not have good project management, these human factors

can lead to decision fatigue and resource limits that might prevent us from finishing the application on time.

1.8.2 User Adoption and Digital Workflow Transition

Notable introduces a new second brain architecture that requires students to shift from familiar but fragmented tools. The primary risk is user adoption, as many ignore the biological tax and focus depletion caused by constant context switching. To mitigate this, we will ensure the interface is intuitive and provides the seamless experience needed for widespread community adoption

1.8.3 Configuration Trap

Notable must avoid the configuration trap where users spend 2 to 10 hours setting up their software instead of doing their actual work. When tools are fragmented, it leads to data decay and administrative entropy, meaning information becomes generic or disconnected across different databases. This can also cause a stateless reasoning flaw where the application has no memory of the student's broader context or history. All of these issues contribute to tool fatigue, making the software feel exhausting to use.

1.8.4 Competition Risk

We recognize that similar productivity applications exist, but they often act as disconnected functional silos that increase technical debt and tool fatigue. Many current services are limited by regional data or restrictive monetization that locks essential academic features behind paywalls.

CHAPTER II
PROJECT MANAGEMENT

2.1 Project Scope & Time Frame

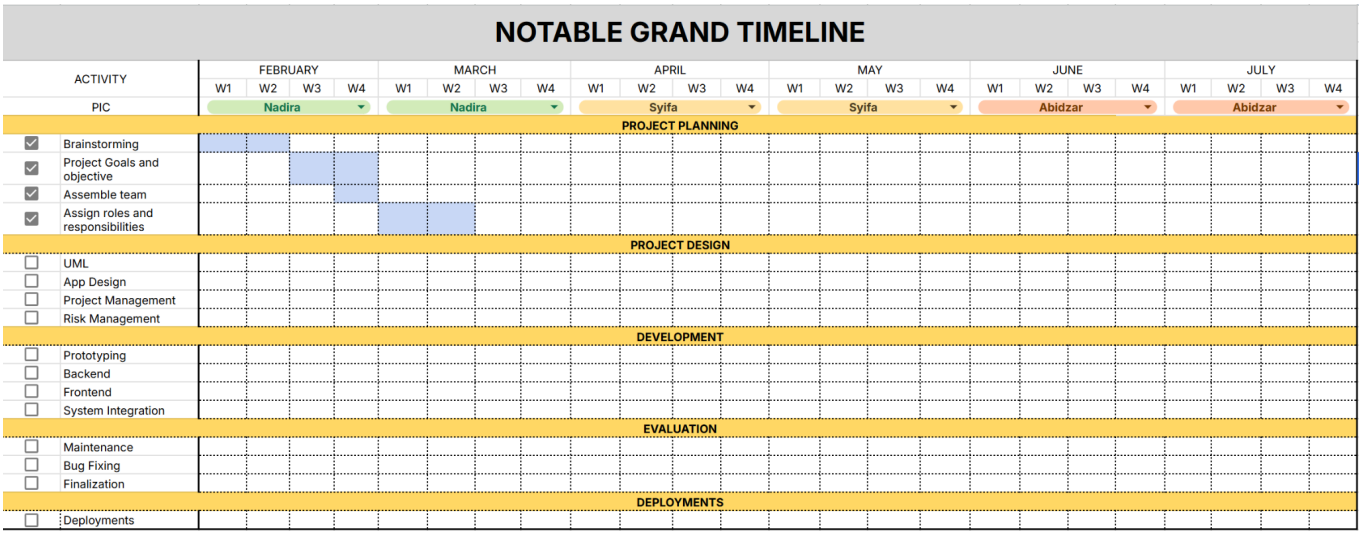
2.1.1 Project Scope

The Notable project is based on creating a unified interface for a productivity application that is meant to assist students in their academic endeavors without having to deal with the hassle of switching between various applications. The scope of this project is based on integrating a quick list system with a note system into one interface. This is aimed at eliminating the toggle tax, which is four hours of productivity that students lose every week simply due to switching applications.

This project is limited to a single academic semester, with a focus on "radical simplicity". We want to guarantee that a student can configure their entire workspace in under ten minutes, which directly addresses the problem of the configuration trap, which is the waste of hours just making a system work. Unlike many other tools, **Notable will offer features like reminders and deadlines for free, so every student can use basic organization tools.**

Notable aims to give students free access to essential study tools like reminders and scheduling, reducing financial barriers to staying organized. It also eliminates the lengthy setup process typical of productivity software, getting students fully set up in under ten minutes instead of hours, so they can focus on studying rather than configuring their tools.

2.1.2 Time Frame



2.2 Development Process Identification

2.2.1 Methods

- AI Orchestration (Vibe Coding): The project leverages "vibe coding" for rapid functional prototyping, allowing the team to implement complex features such as state management and initial UI scaffolding, in minutes rather than weeks This process relies on high-quality natural language prompts to guide AI agents in autonomous codebase exploration and proactive debugging
- Human-Centric Engineering: While AI generates syntax, human developers focus on critical evaluation and strategic decisions The process adheres to the Model-View-Controller (MVC) paradigm to ensure modularity and scalability
- Professional Standards: The software development lifecycle (SDLC) is aligned with IEEE 12207-2017 standards, ensuring a systematic transition from requirements to final evaluation

2.2.2 Tools

- **VSCode**

Serves as the central command center for both manual coding and "vibe coding" sessions, leveraging AI extensions to accelerate development

- **Neon (PostgreSQL) and SQLite (The Dual-Data Strategy):**
    - SQLite is used for local-first on-device storage, ensuring the app feels "snappy" (like Todoist) and functions offline
    - Neon (PostgreSQL) provides a serverless cloud backend. Its serverless nature offers a robust free tier, making it possible to scale the app without the high costs that usually require a subscription model
- **GitHub**

  Used for complete code ownership and version control. It enables automated testing via GitHub Actions to ensure that AI-generated updates do not disrupt existing system stability
- **Figma**

  The primary design tool for mapping out an inclusive interface. Figma allows the team to apply physical ADA reach-range standards to virtual elements, ensuring that interactive targets stay within the optimal 15 to 48-inch range for all users.

2.2.3 Language

- TypeScript/JavaScript: The core programming languages for the web-based frontend and backend, providing a modern, type-safe environment that is highly compatible with AI coding agents

- Natural Language: Precise English serves as a high-level interface, where the developer’s intent is translated into executable logic by AI tools

2.3 Organizational Structure

The Notable project is managed on a rotating Person-in-Charge (PIC) basis, with each team member taking ownership of the project for a two-month period. This ensures distributed responsibility and accountability across the project lifecycle.

- Nadira serves as PIC from February to March, overseeing the Project Planning phase, including brainstorming, goal-setting, team assembly, and role assignment.
- Syifa serves as PIC from April to May, overseeing the Project Design and early Development phases, including UML, app design, project management, risk management, and prototyping.
- Abidzar serves as PIC from June to July, overseeing the latter Development, Evaluation, and Deployment phases, including backend, frontend, system integration, bug fixing, finalization, and deployment.

This rotating structure ensures that each team member develops leadership experience while maintaining continuity throughout the project. **However, it is to be noted that every team member will directly handle every process of coding and the developing of the project.**

2.4 Risk Management Overview

The risk management process for the Notable application such as Implementation Failure, Mental Fatigue, Timeline Limitation, Forgotten Urgency, and Force Majeure. These are potential obstacles to the achievement of the goal of reducing toggle tax and cognitive load for BHPS prioritization reliability and integration smoothness.

- Implementation Failure (Priority 1, highest impact) covers important technical risks such as module integration issues (e.g., the inability of the BHPS to update when tasks are edited), untrustworthy AI-coded software produced by the orchestration/vibe coding approach, and AI service vendor updates. Mitigation

strategies include unit testing of modules, separation of concerns using the MVC pattern, "fail fast" prototyping, and manual code reviews prior to integration into the master branch.

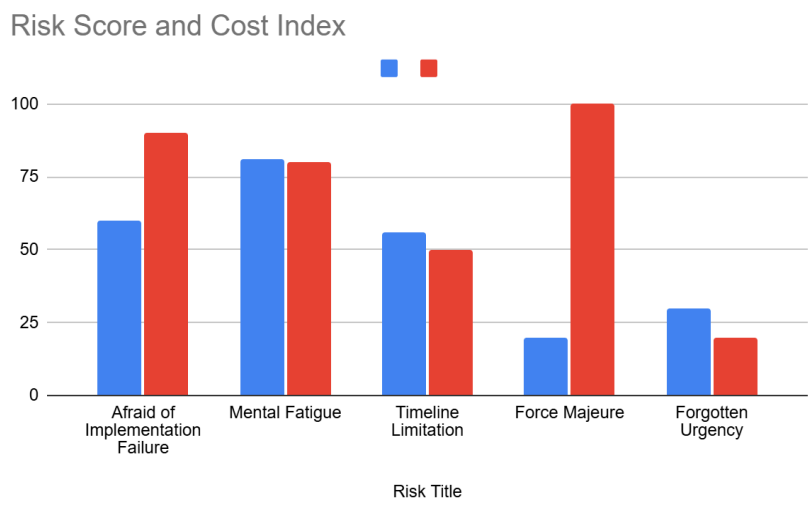
- Mental Fatigue (Priority 2) concerns the burnout risk associated with managing student academics alongside the development of features such as task capture. Countermeasures include breaks during sprints, task rotation, and GTD-based workflows similar to the benefits provided by Notable.

- Timeline Limitation (Priority 3) concerns the semester timeline through the development of the MVP. This includes the development of "must-haves" such as free reminders and 10-minute setups, as well as the integration of a functional BHPS, task search, and focus-preserving modules.

- Forgotten Urgency (Priority 4) ensures deadlines are met for Todoist/Notion hybrids through the integration of Jira/Trello reminders and 10-minute Monday syncs.

- Force Majeure (Priority 5, lowest likelihood) concerns cloud file redundancy and remote working. Additional strategies include weekly meetings to avoid the "configuration trap" experienced by Notion users, the development of Notable-style task boards to instill trust in BHPS suggestions, and the use of peer testing to instill trust in the software.

RISK ANALYSIS AND MITIGATION PLAN								
No.	Risk Title	Likelihood (1-10)	Impact (1-10), 1=least impact	Retirement Cost (1-10), 1=lowest cost	Priority (Rank), 1=Highest Priority	Risk Score	Cost Index	Retirement / Mitigation Plan
1	Afraid of Implementation Failure	6	10	9	1	60	90	Implementation of modular unit testing and a "fail-fast" prototyping phase.
2	Mental Fatigue	9	9	8	2	81	80	Scheduled "sprint breaks" and task rotation to prevent developer burnout.
3	Timeline Limitation	7	8	5	3	56	50	Identification of "must-have" vs "nice-to-have" features; prioritize MVP.
4	Force Majeure	2	10	10	5	20	100	Cloud-based redundancy for all project files and remote work protocol.
5	Forgotten Urgency	5	6	2	4	30	20	Automated Jira/Trello reminders and a dedicated 10-minute Monday sync.



NOTABLE

The Risk Score is calculated from Likelihood × Impact (standard risk management formula), example:

- Mental Fatigue: 9 × 9 = 81 → very high likelihood (9/10), high impact (9/10), hence the highest score.
- Force Majeure: 2 × 10 = 20 → extreme impact (10/10) but very low likelihood (2/10), hence the low score.

Whilst cost index was actually just a scale conversion from a 1–10 scale to a 0–100 scale. The formula is: (value / max value) × 100, example:

- Force Majeure: Retirement Cost = 10 → 10/10(100) = 100 (most expensive to mitigate)
- Forgotten Urgency: Retirement Cost = 2 → 2/10(100) = 20 (cheapest to mitigate)

Notable's Risk Analysis and Mitigation Plan identifies and quantifies the key risks that may prevent the application from delivering its promised benefits of reduced "toggle tax" and cognitive load. The highest-priority risk is Afraid of Implementation Failure (Risk Score: 60, Priority 1), covering poor module integration, unreliable AI-generated code, and potential vendor switches. Mental Fatigue holds the highest overall risk score (81, Priority 2) due to its high likelihood, reflecting the danger of Notable falling into the same "configuration trap" it aims to solve. Forgotten Urgency (Priority 4) captures the risk of students disregarding BHPS suggestions, while Timeline Limitation (Priority 3) and Force Majeure (Priority 5) represent scheduling and external disruption risks respectively.

To mitigate these, the team will apply incremental development, the MVC pattern, manual code reviews, weekly meetings, and milestone based tracking tied to the BHPS, task capture, and study module deliverables.

2.5 Documents to be Produced

To maintain a single source of truth and mitigate the risks of data decay or administrative entropy, the Notable project will produce a standardized set of technical and managerial documents. These documents are intended to serve as the project's external memory, such that all information is interconnected. This prevents the stateless reasoning flaws associated with a loss of context.

- **Memorandum of Understanding (MOU)**

  The MOU is a foundational agreement to manage human capital and the multidisciplinary roles of the team. By explicitly defining the responsibilities of project management, development, and design, the MOU mitigates the risks associated with collaborative working, such as difficulties in communication or

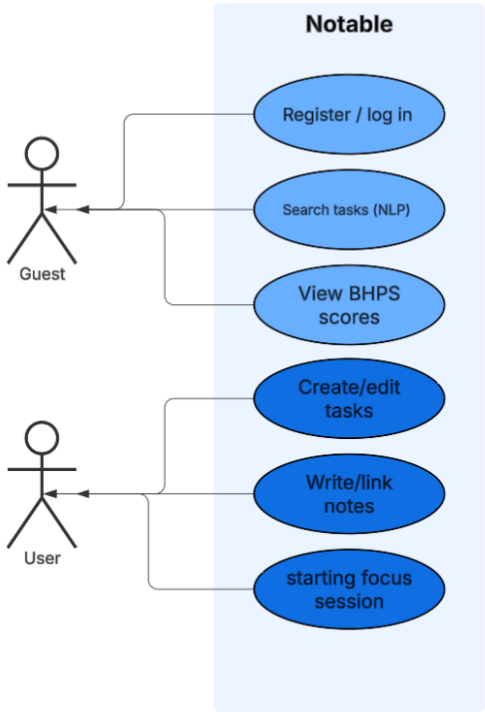
the definition of roles. It is a strategic mandate to protect the team's flow state and maintain the focus of the human resources on high-level strategic thinking rather than friction.

- **Final Project Management Report**

  The final report integrates theoretical and practical knowledge by briefly describing the development process and lessons learned during the semester.

CHAPTER III
DESIGN DIAGRAMS

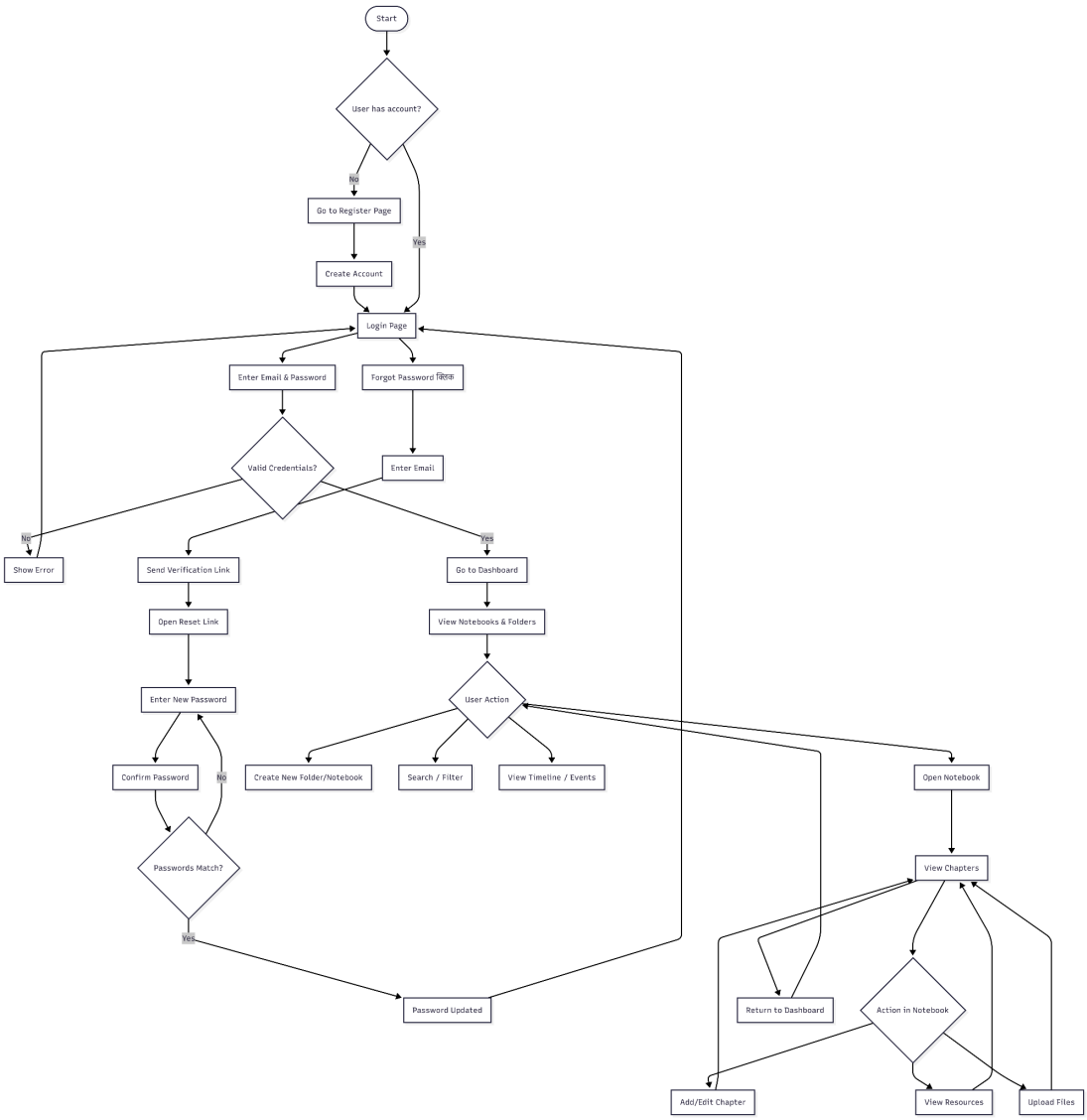
3.1 Use Case Diagram



The use case diagram above represents the features that can be done in the Notable app. Three types of actors can interact with the Notable app:

- Guest: A user that is not logged in. They can only access the registration and login page to create an account or sign in. Once authenticated, they transition into a logged-in User with full access to the platform's productivity features.
- User (Logged In): A user that is logged in and has an active account. The User role is the primary actor of Notable here they are able to create notes, folders, to-do’s and start focus sessions, and nearly all core productivity features are designed around their daily academic workflow.

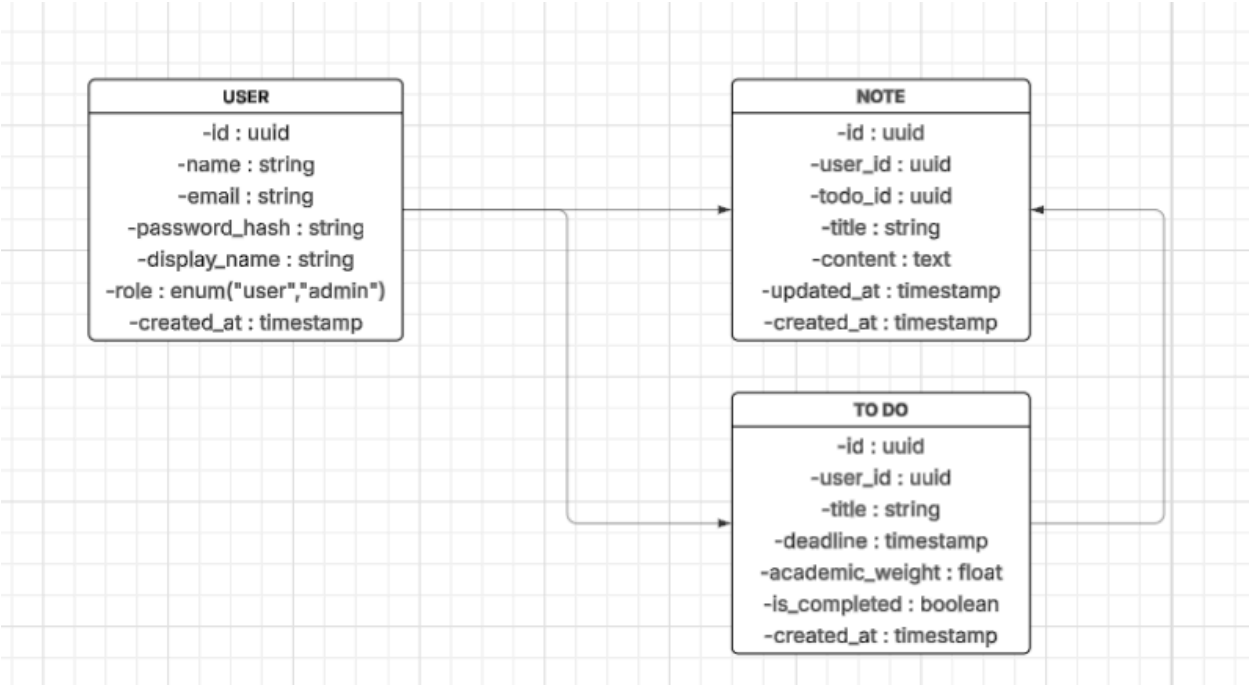
3.2 Activity Diagram



This is an activity diagram that shows the main user flow through the application from accessing their account to using the main features. A user first determines whether they have an account or not. If they do not have an account, they are supposed to register, then they are directed to the login page. For a user who already has an account, they are supposed to login using their details, which are then validated to allow them to access their dashboard. If a user fails to login, they are supposed to try again or click on “Forgot Password” to request a verification email, reset their password, then login.

After a successful login, a user is then taken to their dashboard where they can view, create, search, or manage their notebooks, folders, as well as their own timeline. When a user clicks on a notebook, they are then directed to where they can view or manage their chapters, such as adding or editing content, as well as resources. A user can then go back to their dashboard to continue using the system.

3.3 Object Diagram



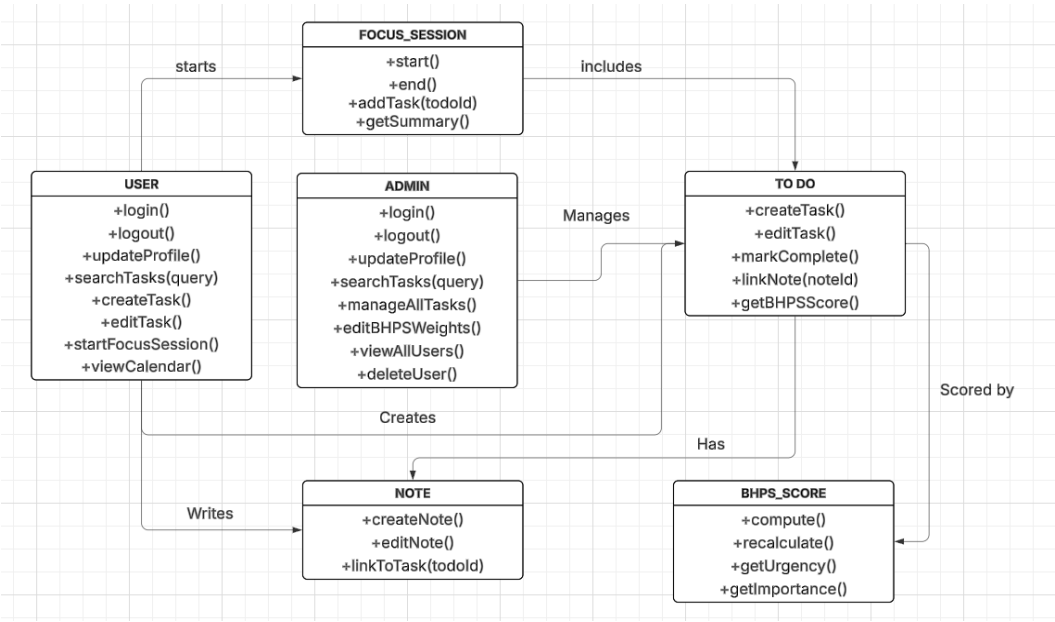
The object diagram we created consists of three main objects that will be used:USER, TODO, and NOTE.

USER stores the core data of each Notable user, including their name, email, password hash, display name, role, and account creation timestamp. It also records the user's role, whether as a regular user with standard productivity access or an admin who has broader rights to manage the platform and oversee all tasks across the system.

TODO is the central object structure that users will create and interact with daily. Each TODO connects back to both the USER who wrote it and optionally the NOTE it belongs to. Each TODO entry has attributes such as title, deadline, completion status, and creation timestamp. This information helps users organize their workload, reduce cognitive load, and avoid missing high-priority academic deadlines.

NOTE is the central object structure that users will create and interact with daily. is an object structure that stores written content linked to a specific task or created as a standalone entry. through foreign key references (user_id and todo_id). It records the note's title, full text content, and both creation and last-updated timestamps. With NOTE, users can capture detailed context, references, or reflections alongside their tasks, turning Notable into a true second-brain system rather than just a simple to-do list.

3.4 Class Diagram



Based on the Class Diagram provided, we plan to create six main classes:

USER, ADMIN, TODO, NOTE, BHPS_SCORE, and FOCUS_SESSION. Although USER and ADMIN share several similarities, we consider making them separate classes due to differences in authority and system access. Both classes, USER and ADMIN, can perform authentication features such as login and logout, update their profile, and search through tasks. However, what makes them different is that ADMIN can manage all tasks across the entire platform, adjust the BHPS scoring weights, and oversee all registered users — privileges that a regular USER does not have.

Since our focus is on reducing cognitive load and helping students prioritize their academic workload effectively, the TODO class serves as the central class of the system. It holds the core task data and provides methods to create, edit, and mark tasks as complete, as well as link associated notes and retrieve the computed BHPS score for that task.

The NOTE class works in close relationship with TODO, allowing users to attach written context, references, or reflections to any given task. This transforms Notable from a simple checklist into a deeper knowledge management tool.

The BHPS_SCORE class is the analytical engine behind Notable's smart prioritization. It computes and recalculates each task's priority score based on urgency, academic importance, and estimated effort, providing the system with a dynamic ranking that updates as the student's workload evolves.

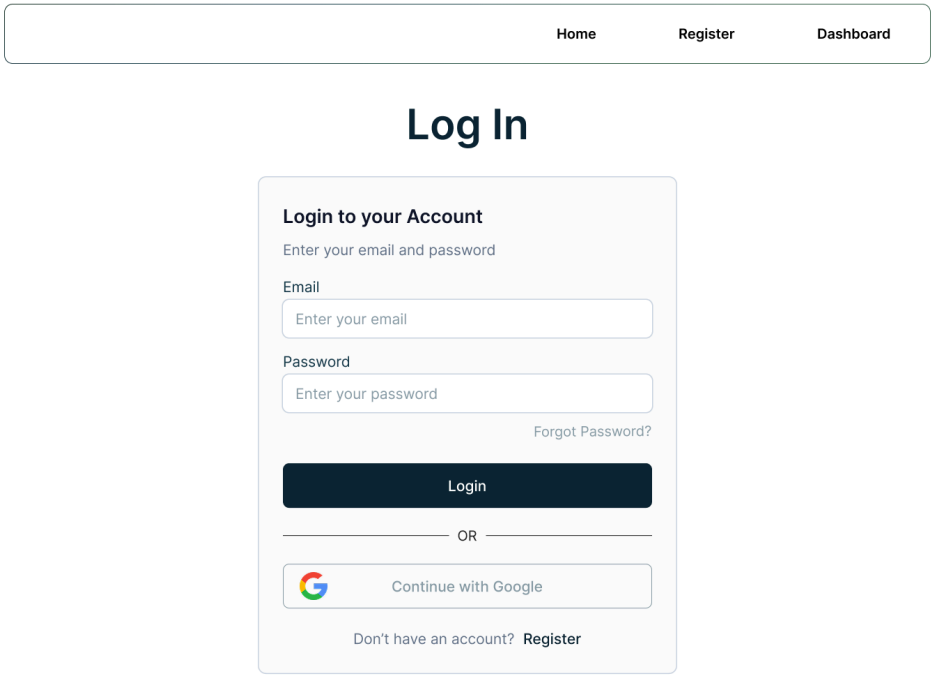
Finally, the FOCUS_SESSION class manages the structured study sessions that Notable recommends to users. It allows users to start and end a timed session, group related tasks together within that session, and retrieve a summary upon completion — directly supporting the goal of protecting the student's flow state and minimizing the toggle tax throughout their study period.

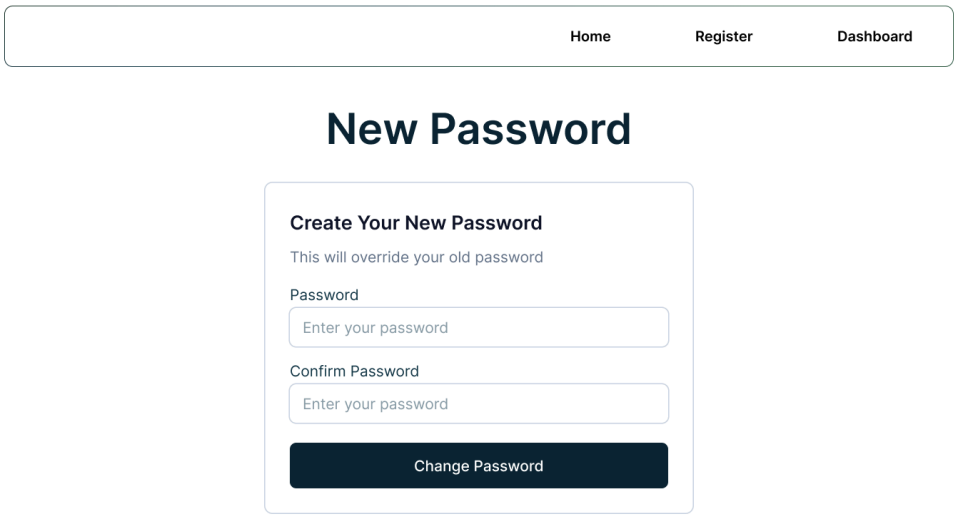
3.5 Wireframe

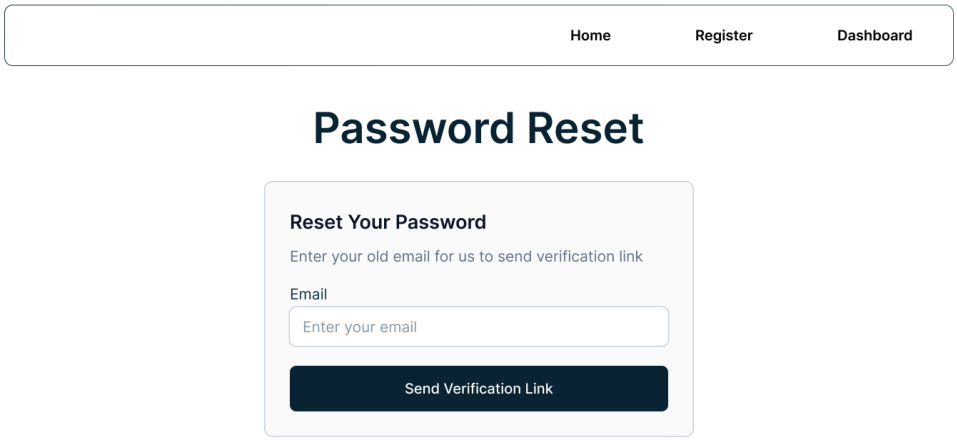
The application has several main pages, which are considered to be a core experience. There are two sets of pages. The first set includes authentication pages, which are Log In, Sign Up, and Reset Password pages. The second set includes Dashboard and Notebook View pages. In these pages, users will be able to interact with their content.

The pages in the application use a consistent design system. The font used in the application includes Inria Sans and Inter. The design of the pages focuses on providing a clear and intuitive experience. The design of all pages in the application includes a consistent style, which helps in providing a better user experience.

3.5.1 Authentication Pages

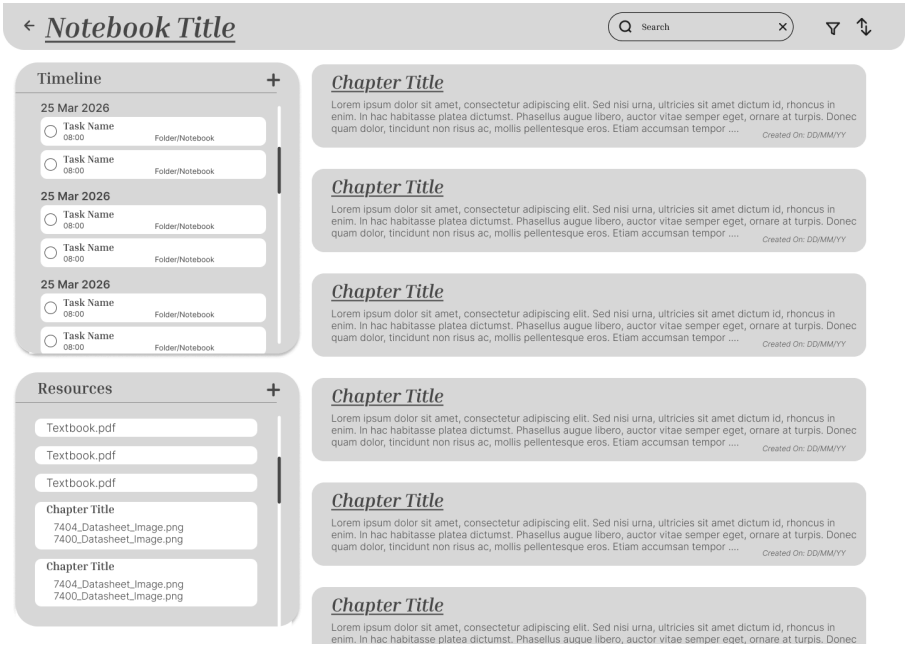



Home
Register
Dashboard
New Password
Create Your New Password
This will override your old password
Password
Enter your password
Confirm Password
Enter your password
Change Password


Home
Register
Dashboard
Password Reset
Reset Your Password
Enter your old email for us to send verification link
Email
Enter your email
Send Verification Link

3.5.2 Dashboard

3.5.3 Notebook View Page



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